Rapport d'activité

Campagne r2SCAN d'Al_4C_3 : méthodologie, structures d'entrée et maillage **k**, en vue de la comparaison avec le scan PBE

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Résumé

Ce rapport documente le lancement de la campagne r2SCAN d'Al_4C_3, destinée à être comparée au scan PBE complet (`rapport_activite_Al4C3_HP_scan.tex`, 7 polymorphes, 0–100 GPa). Les paramètres DFT (`ENCUT`, `EDIFF`, `KSPACING`, la grille de pression, `ISIF=8`) sont identiques à ceux du scan PBE ; seule la fonctionnelle change (PBE → r2SCAN), pour une comparaison à un seul paramètre variable. Chaque point r2SCAN est amorcé depuis le `WAVECAR`/`CHGCAR` PBE déjà convergé à la même pression (pratique recommandée par VASP pour les meta-GGA). Ce document liste, pour chacun des sept polymorphes convergés en PBE, la structure d'entrée (`POSCAR`, identique à 0 et 50 GPa entre PBE et r2SCAN) et le maillage de points **k** généré par `KSPACING=0.2` à ces deux pressions. Il ne contient *pas encore* les résultats r2SCAN eux-mêmes (H(P), classement de stabilité) : les chaînes de calcul viennent d'être soumises (voir section 5) et ce rapport sera complété d'une comparaison quantitative PBE/r2SCAN une fois les points convergés.

1 Contexte et objectifs

Le scan PBE (0–100 GPa, sept polymorphes, `KSPACING=0.2`) a identifié une séquence de stabilité Pnma-VII → Pbam → Pnma avec deux transitions à 26 et 52 GPa (voir rapport principal, section « Précision des pressions de transition », pour l'incertitude associée à ces valeurs). L'objectif de la présente campagne est de vérifier que cette séquence – et en particulier les deux pressions de transition – est robuste au choix de fonctionnelle d'échange-corrélation : r2SCAN (meta-GGA) corrige une partie des biais systématiques connus de PBE (notamment la sur-estimation des volumes d'équilibre), ce qui peut déplacer les croisements d'enthalpie entre phases de volumes différents (cf. l'analyse de précision du rapport principal, où $\delta P_t \propto 1/|\Delta V|$).

2 Méthodologie

2.1 Paramètres DFT et redémarrage depuis PBE

Les paramètres communs (`PREC`, `ENCUT=600 eV`, `EDIFF=1E-5`, `ISMEAR/SIGMA=0/0.05`, `KSPACING=0.2`, `IBRION/ISIF=2/8`, `NSW=200`, `EDIFFG=-0.01`, grille de pression 0–100 GPa par pas de 10 GPa, `KPAR=8`) sont repris à l'identique du scan PBE. Les seuls ajouts sont les balises meta-GGA :

Balise	Rôle
`METAGGA = R2scan`	fonctionnelle meta-GGA r2SCAN (implémentation native VASP)
`LASPH = .TRUE.`	corrections PAW asphériques (nécessaires en meta-GGA)
`ADDGRID = .TRUE.`	grille réelle plus fine pour les termes de gradient de τ
`LMIXTAU = .TRUE.`	mélange de la densité d'énergie cinétique dans le SCF

Pour les sept polymorphes dont le scan PBE est intégralement convergé (0–100 GPa), chaque point r2SCAN est amorcé *directement* depuis le `CONTCAR`/`WAVECAR`/`CHGCAR` PBE convergé à la même pression (`ISTART=1`, `ICHARG=1`) – c'est la pratique recommandée par VASP pour la convergence SCF des meta-GGA, et cela élimine le besoin d'une chaîne de redémarrage entre pressions voisines (chaque point est indépendant). Pour le polymorphe **Fd-3m**, dont le scan PBE est encore en cours (voir section 5), seul le point $P = 0$ dispose d'un WAVECAR PBE convergé ; la chaîne r2SCAN de ce polymorphe est donc construite comme une chaîne ascendante unique $0 \to 10 \to \cdots \to 100$ GPa, où seul $P = 0$ est amorcé depuis PBE et chaque point suivant redémarre depuis le point r2SCAN précédent (même mécanisme que la chaîne PBE originale de Fd-3m).

3 Maillage de points k

Le maillage KSPACING=0.2 génère un nombre de points **k** différent pour chaque polymorphe (la maille et le réseau réciproque diffèrent), et identique entre PBE et r2SCAN à pression donnée (seule la fonctionnelle change, pas la géométrie du maillage). Le tableau ci-dessous liste le maillage de Monkhorst-Pack généré ($N_1 \times N_2 \times N_3$), le nombre total de points **k** et le nombre de points irréductibles (après réduction par symétrie) à 0 et 50 GPa, extraits des OUTCAR PBE convergés.

Polymorphe	P (GPa)	Maillage $N_1 \times N_2 \times N_3$	k-points totaux	k-points irréductibles
Pnma-VII	0	$4 \times 10 \times 4$	160	54
Pnma-VII	50	$4 \times 11 \times 4$	176	54
Pbam	0	$4 \times 7 \times 11$	308	72
Pbam	50	$4 \times 7 \times 11$	308	72
Pnma	0	$4 \times 11 \times 4$	176	54
Pnma	50	$4 \times 11 \times 4$	176	54
R-3m	0	$12 \times 12 \times 2$	288	38
R-3m	50	$12 \times 12 \times 2$	288	38
Pnma-III	0	$4 \times 11 \times 4$	176	54
Pnma-III	50	$4 \times 11 \times 4$	176	54
anti-$CaFe_2O_4$	0	$11 \times 4 \times 4$	176	54
anti-$CaFe_2O_4$	50	$12 \times 4 \times 4$	192	63
I-43d	0	$5 \times 5 \times 5$	125	10
I-43d	50	$6 \times 6 \times 6$	216	20

Le nombre de points **k** irréductibles varie fortement d'un polymorphe à l'autre (de 10 pour I-43d à 72 pour Pbam), reflétant à la fois des mailles de tailles différentes (14 à 28 atomes/maille selon le polymorphe) et des symétries de groupe d'espace différentes. I-43d se distingue par un maillage cubique dense (5^3 à 6^3) mais très peu de points irréductibles (groupe d'espace cubique $I\bar{4}3d$, haute symétrie) ; à l'inverse R-3m, malgré 288 points totaux, n'en réduit qu'à 38 (symétrie rhomboédrique moins favorable pour ce maillage anisotrope).

4 Structures d'entrée (POSCAR, PBE relaxé → seed r2SCAN)

Les structures ci-dessous sont les CONTCAR PBE convergés à 0 et 50 GPa, utilisés tels quels comme POSCAR d'entrée des calculs r2SCAN correspondants (hp_curves_r2SCAN/Al4C3_<polymorphe>/P<press

Pnma-VII – 0 GPa

```
Al4C3 Pnma (CIF findsym, P21/n21/m21/a)
   1.00000000000000
     9.6447110324068994    0.0000000000000000    0.0000000000000000
```

```
     0.0000000000000000    3.3137960964664095    0.0000000000000000
     0.0000000000000000    0.0000000000000000   10.1672966455862142
   Al   C
    16    12
Direct
  0.2638924603987636  0.2500000000000000  0.2677086017912220
  0.7638924603987637  0.2500000000000000  0.2322913982087779
  0.7361075396012363  0.7500000000000000  0.7322913982087779
  0.2361075396012364  0.7500000000000000  0.7677086017912221
  0.4687336063193806  0.2500000000000000  0.8920766296137922
  0.9687336063193805  0.2500000000000000  0.6079233703862078
  0.5312663936806195  0.7500000000000000  0.1079233703862078
  0.0312663936806194  0.7500000000000000  0.3920766296137922
  0.5078696080686969  0.2500000000000000  0.6025813718867934
  0.0078696080686969  0.2500000000000000  0.8974186281132066
  0.4921303919313030  0.7500000000000000  0.3974186281132066
  0.9921303919313031  0.7500000000000000  0.1025813718867935
  0.2411735741243224  0.2500000000000000  0.5605001914892204
  0.7411735741243224  0.2500000000000000  0.9394998085107796
  0.7588264258756776  0.7500000000000000  0.4394998085107796
  0.2588264258756777  0.7500000000000000  0.0605001914892204
  0.3845072240206697  0.2500000000000000  0.4241631201653526
  0.8845072240206696  0.2500000000000000  0.0758368798346473
  0.6154927759793304  0.7500000000000000  0.5758368798346473
  0.1154927759793303  0.7500000000000000  0.9241631201653527
  0.3904520877040922  0.2500000000000000  0.0833557375444348
  0.8904520877040921  0.2500000000000000  0.4166442624555652
  0.6095479122959079  0.7500000000000000  0.9166442624555650
  0.1095479122959078  0.7500000000000000  0.5833557375444350
  0.3496738716245690  0.2500000000000000  0.7346401141650764
  0.8496738716245689  0.2500000000000000  0.7653598858349236
  0.6503261283754311  0.7500000000000000  0.2653598858349236
  0.1503261283754311  0.7500000000000000  0.2346401141650764

  0.00000000E+00  0.00000000E+00  0.00000000E+00
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  0.00000000E+00  0.00000000E+00  0.00000000E+00
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  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

Pnma-VII – 50 GPa

```
Al4C3 Pnma (CIF findsym, P21/n21/m21/a)
   1.00000000000000
     8.9923981068448793    0.0000000000000000    0.0000000000000000
     0.0000000000000000    3.0896699387060820    0.0000000000000000
     0.0000000000000000    0.0000000000000000    9.4796390270578463
   Al   C
    16    12
Direct
  0.2681833644661538  0.2500000000000000  0.2664767518332596
  0.7681833644661538  0.2500000000000000  0.2335232481667404
  0.7318166355338462  0.7500000000000000  0.7335232481667404
  0.2318166355338462  0.7500000000000000  0.7664767518332596
  0.4656894866399242  0.2500000000000000  0.8916382141646366
  0.9656894866399242  0.2500000000000000  0.6083617858353634
  0.5343105133600758  0.7500000000000000  0.1083617858353634
  0.0343105133600758  0.7500000000000000  0.3916382141646366
  0.5060443207492540  0.2500000000000000  0.6048072554787396
  0.0060443207492540  0.2500000000000000  0.8951927445212604
  0.4939556792507460  0.7500000000000000  0.3951927445212604
  0.9939556792507460  0.7500000000000000  0.1048072554787396
  0.2367900995859671  0.2500000000000000  0.5579400534326382
  0.7367900995859671  0.2500000000000000  0.9420599465673618
  0.7632099004140329  0.7500000000000000  0.4420599465673618
  0.2632099004140329  0.7500000000000000  0.0579400534326382
  0.3846996388742596  0.2500000000000000  0.4253344150536265
  0.8846996388742596  0.2500000000000000  0.0746655849463735
  0.6153003611257404  0.7500000000000000  0.5746655849463735
  0.1153003611257404  0.7500000000000000  0.9253344150536265
  0.3944047199639513  0.2500000000000000  0.0877513538253609
  0.8944047199639513  0.2500000000000000  0.4122486461746391
  0.6055952800360487  0.7500000000000000  0.9122486461746391
  0.1055952800360487  0.7500000000000000  0.5877513538253609
  0.3471076641912347  0.2500000000000000  0.7342183592237674
  0.8471076641912347  0.2500000000000000  0.7657816407762326
  0.6528923358087653  0.7500000000000000  0.2657816407762326
  0.1528923358087653  0.7500000000000000  0.2342183592237674

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```

Pbam – 0 GPa

```
Al4C3 Pbam polymorph (findsym-output) -
   1.00000000000000
     9.0566422846240506    0.0000000000000000    0.0000000000000000
     0.0000000000000000    5.2293835261696859    0.0000000000000000
     0.0000000000000000    0.0000000000000000    3.0977367561939881
   Al   C
     8     6
Direct
  0.7846720435396047  0.1531142601818623 -0.0000000000000000
  0.2153279564603954  0.8468857398181378  0.0000000000000000
  0.7153279564603953  0.6531142601818622 -0.0000000000000000
  0.2846720435396045  0.3468857398181377  0.0000000000000000
  0.0250541557709346  0.2542515312388905  0.5000000000000000
  0.9749458682290674  0.7457484687611025  0.5000000000000000
  0.4749458382290650  0.7542515312388975  0.5000000000000000
  0.5250541317709326  0.2457484687611024  0.5000000000000000
  0.0000000000000000 -0.0000000000000000  0.0000000000000000
  0.5000000000000000  0.5000000000000000  0.0000000000000000
  0.8272803098748985  0.4538044740481856  0.5000000000000000
  0.1727196901251014  0.5461955259518143  0.5000000000000000
  0.6727196901251015  0.9538044740481857  0.5000000000000000
  0.3272803098748985  0.0461955259518144  0.5000000000000000

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```

Pbam – 50 GPa

```
Al4C3 Pbam polymorph (findsym-output) -
   1.00000000000000
     8.4823493958000000    0.0000000000000000    0.0000000000000000
     0.0000000000000000    4.8977818489000002    0.0000000000000000
     0.0000000000000000    0.0000000000000000    2.9013054371000000
   Al   C
     8     6
Direct
  0.7858800289999976  0.1601099969999993  0.0000000000000000
  0.2141199710000024  0.8398900030000007  0.0000000000000000
  0.7141199710000024  0.6601099969999993  0.0000000000000000
  0.2858800289999976  0.3398900030000007  0.0000000000000000
  0.0310999989999985  0.2530300020000027  0.5000000000000000
  0.9689000250000035  0.7469699979999973  0.5000000000000000
  0.4688999950000010  0.7530300020000027  0.5000000000000000
  0.5310999749999965  0.2469699979999973  0.5000000000000000
  0.0000000000000000  0.0000000000000000  0.0000000000000000
```

```
  0.5000000000000000  0.5000000000000000  0.0000000000000000
  0.8245900269999993  0.4467200039999994  0.5000000000000000
  0.1754099730000007  0.5532799960000006  0.5000000000000000
  0.6754099730000007  0.9467200039999994  0.5000000000000000
  0.3245900269999993  0.0532799960000006  0.5000000000000000

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  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

Pnma – 0 GPa

```
Al4C3 Pnma (POSCAR.IV, symmetrized via s
   1.00000000000000
     8.8687877986054335    0.0000000000000000    0.0000000000000000
     0.0000000000000000    3.0595706192262035    0.0000000000000000
     0.0000000000000000    0.0000000000000000   10.5424912694858808
   Al   C
    16    12
Direct
  0.2735502672922275  0.2500000000000000  0.3075329915271308
  0.2264497327077724  0.7500000000000000  0.8075329915271308
  0.7264497327077724  0.7500000000000000  0.6924670084728692
  0.7735502672922276  0.2500000000000000  0.1924670084728693
  0.3615341662279118  0.2500000000000000  0.0187587654397753
  0.1384658337720882  0.7500000000000000  0.5187587654397752
  0.6384658337720881  0.7500000000000000  0.9812412345602248
  0.8615341662279119  0.2500000000000000  0.4812412345602247
  0.5160590391368277  0.2500000000000000  0.7948720507037115
  0.9839409608631723  0.7500000000000000  0.2948720507037115
  0.4839409608631723  0.7500000000000000  0.2051279492962884
  0.0160590391368277  0.2500000000000000  0.7051279492962885
  0.0813324512462430  0.2500000000000000  0.0833467341491087
  0.4186675487537640  0.7500000000000000  0.5833467341491085
  0.9186675487537570  0.7500000000000000  0.9166532658508915
  0.5813324512462430  0.2500000000000000  0.4166532658508914
  0.2457612108131275  0.2500000000000000  0.6385376537495265
  0.2542387891868724  0.7500000000000000  0.1385376537495265
  0.7542387891868727  0.7500000000000000  0.3614623462504735
  0.7457612108131273  0.2500000000000000  0.8614623462504735
  0.5875004958432158  0.2500000000000000  0.6136101721742773
  0.9124995041567842  0.7500000000000000  0.1136101721742773
  0.4124995041567915  0.7500000000000000  0.3863898278257227
  0.0875004958432156  0.2500000000000000  0.8863898278257227
  0.0722337674812502  0.2500000000000000  0.3872433425577897
  0.4277662325187498  0.7500000000000000  0.8872433425577972
  0.9277662325187502  0.7500000000000000  0.6127566574422028
  0.5722337674812498  0.2500000000000000  0.1127566574422030

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```

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  0.00000000E+00  0.00000000E+00  0.00000000E+00
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  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

Pnma – 50 GPa

```
Al4C3 Pnma (POSCAR.IV, symmetrized via s
   1.00000000000000
     8.2890294394235049    0.0000000000000000    0.0000000000000000
     0.0000000000000000    2.8595645211794394    0.0000000000000000
     0.0000000000000000    0.0000000000000000    9.8533218385690624
   Al   C
    16    12
Direct
  0.2629312655562072  0.2500000000000000  0.2981934154944613
  0.2370687344437928  0.7500000000000000  0.7981934154944613
  0.7370687344437928  0.7500000000000000  0.7018065845055387
  0.7629312655562072  0.2500000000000000  0.2018065845055387
  0.3469030433064250  0.2500000000000000  0.0112167087473821
  0.1530969566935750  0.7500000000000000  0.5112167087473821
  0.6530969566935749  0.7500000000000000  0.9887832912526179
  0.8469030433064251  0.2500000000000000  0.4887832912526179
  0.5225009079149764  0.2500000000000000  0.7942536374118344
  0.9774990920850236  0.7500000000000000  0.2942536374118344
  0.4774990920850237  0.7500000000000000  0.2057463625881656
  0.0225009079149763  0.2500000000000000  0.7057463625881656
  0.0689590485967116  0.2500000000000000  0.0894805905942223
  0.4310409514032955  0.7500000000000000  0.5894805905942223
  0.9310409514032884  0.7500000000000000  0.9105194094057777
  0.5689590485967116  0.2500000000000000  0.4105194094057777
  0.2592235199889221  0.2500000000000000  0.6419226335859984
  0.2407764800110779  0.7500000000000000  0.1419226335859984
  0.7407764800110779  0.7500000000000000  0.3580773664140016
  0.7592235199889221  0.2500000000000000  0.8580773664140016
  0.6043531701401923  0.2500000000000000  0.6081829935710156
  0.8956468298598077  0.7500000000000000  0.1081829935710156
  0.3956468298598077  0.7500000000000000  0.3918170064289844
  0.1043531701401923  0.2500000000000000  0.8918170064289844
  0.0641760130956121  0.2500000000000000  0.3891272411387470
  0.4358239869043878  0.7500000000000000  0.8891272411387470
  0.9358239869043878  0.7500000000000000  0.6108727588612530
```

```
  0.5641760130956122  0.2500000000000000  0.1108727588612530

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

R-3m – 0 GPa

```
Al4C3 R-3m polymorph (findsym-output, fr
   1.00000000000000
     3.3554638352239903    0.0000000000000000    0.0000000000000000
    -1.6777319176119951    2.9059169228212531    0.0000000000000000
     0.0000000000000000    0.0000000000000000   25.0854610045852198
   Al   C
    12     9
Direct
  0.0000000000000000 -0.0000000000000000  0.6298944952465831
 -0.0000000000000000  0.0000000000000000  0.3701055047534167
  0.6666666870000029  0.3333333429999996  0.9632278082465803
  0.6666666870000029  0.3333333429999996  0.7034388187534176
  0.3333333429999996  0.6666666870000029  0.2965611512465801
  0.3333333429999996  0.6666666870000029  0.0367721727534216
  0.0000000000000000 -0.0000000000000000  0.2065187870111717
 -0.0000000000000000  0.0000000000000000  0.7934812129888283
  0.6666666870000029  0.3333333429999996  0.5398521010111724
  0.6666666870000029  0.3333333429999996  0.1268145409888303
  0.3333333429999996  0.6666666870000029  0.8731854740111745
  0.3333333429999996  0.6666666870000029  0.4601478689888250
  0.0000000000000000  0.0000000000000000  0.7167106476763591
  0.0000000000000000 -0.0000000000000000  0.2832893523236410
  0.6666666870000029  0.3333333429999996  0.0500439796763613
  0.6666666870000029  0.3333333429999996  0.6166226653236381
  0.3333333429999996  0.6666666870000029  0.3833773046763594
  0.3333333429999996  0.6666666870000029  0.9499560393236367
  0.0000000000000000 -0.0000000000000000  0.5000000000000000
  0.6666666870000029  0.3333333429999996  0.8333333129999971
  0.3333333429999996  0.6666666870000029  0.1666666719999981
```

```
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

R-3m – 50 GPa

```
Al4C3 R-3m polymorph (findsym-output, fr
   1.00000000000000
     3.1293263730393432    0.0000000000000000    0.0000000000000000
    -1.5646631865196716    2.7100761358194903    0.0000000000000000
     0.0000000000000000    0.0000000000000000   23.3948564360725193
   Al   C
    12     9
Direct
  0.0000000000000000  0.0000000000000000  0.6295050301620364
  0.0000000000000000  0.0000000000000000  0.3704949698379636
  0.6666666870000029  0.3333333429999996  0.9628383431620335
  0.6666666870000029  0.3333333429999996  0.7038282838379644
  0.3333333429999996  0.6666666870000029  0.2961716861620332
  0.3333333429999996  0.6666666870000029  0.0371616378379684
  0.0000000000000000  0.0000000000000000  0.2060826556182036
  0.0000000000000000  0.0000000000000000  0.7939173443817964
  0.6666666870000029  0.3333333429999996  0.5394159696182044
  0.6666666870000029  0.3333333429999996  0.1272506723817983
  0.3333333429999996  0.6666666870000029  0.8727493426182065
  0.3333333429999996  0.6666666870000029  0.4605840003817931
  0.0000000000000000  0.0000000000000000  0.7166862778946808
  0.0000000000000000  0.0000000000000000  0.2833137221053192
  0.6666666870000029  0.3333333429999996  0.0500196098946830
  0.6666666870000029  0.3333333429999996  0.6166470351053164
  0.3333333429999996  0.6666666870000029  0.3833529348946811
  0.3333333429999996  0.6666666870000029  0.9499804091053150
  0.0000000000000000  0.0000000000000000  0.5000000000000000
  0.6666666870000029  0.3333333429999996  0.8333333129999971
  0.3333333429999996  0.6666666870000029  0.1666666719999981

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

```
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

Pnma-III – 0 GPa

```
Al4C3 Pnma (POSCAR.III, symmetrized via
   1.00000000000000
     9.3185950946021627    0.0000000000000000    0.0000000000000000
     0.0000000000000000    3.1424134406869162    0.0000000000000000
     0.0000000000000000    0.0000000000000000   10.1601230062129950
   Al   C
    16    12
Direct
  0.7449744616347906  0.2500000000000000  0.7201669400270182
  0.7550255383652094  0.7500000000000000  0.2201669400270180
  0.2550255383652096  0.7500000000000000  0.2798330599729819
  0.2449744616347904  0.2500000000000000  0.7798330599729818
  0.6082545802170108  0.2500000000000000  0.4786801871503972
  0.8917454197829892  0.7500000000000000  0.9786801871503900
  0.3917454197829891  0.7500000000000000  0.5213198128496100
  0.1082545802170108  0.2500000000000000  0.0213198128496100
  0.1325614982528854  0.2500000000000000  0.5176342081888706
  0.3674385017471147  0.7500000000000000  0.0176342081888705
  0.8674385017471147  0.7500000000000000  0.4823657918111295
  0.6325614982528853  0.2500000000000000  0.9823657918111294
  0.9830998624780699  0.2500000000000000  0.2823658008199360
  0.5169001375219301  0.7500000000000000  0.7823658008199362
  0.0169001375219303  0.7500000000000000  0.7176341991800638
  0.4830998624780697  0.2500000000000000  0.2176341991800640
  0.9307094114206790  0.2500000000000000  0.6227096332100521
  0.5692905885793210  0.7500000000000000  0.1227096332100522
  0.0692905885793140  0.7500000000000000  0.3772903667899479
  0.4307094114206860  0.2500000000000000  0.8772903667899479
  0.3977891753639959  0.2500000000000000  0.3993912080842171
  0.1022108246360040  0.7500000000000000  0.8993912080842100
  0.6022108246360039  0.7500000000000000  0.6006087919157900
  0.8977891753639961  0.2500000000000000  0.1006087919157901
  0.7741899310421310  0.2500000000000000  0.3593595510598142
  0.7258100689578690  0.7500000000000000  0.8593595510598141
  0.2258100689578688  0.7500000000000000  0.6406404489401859
  0.2741899310421312  0.2500000000000000  0.1406404489401859

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

```
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

Pnma-III – 50 GPa

```
Al4C3 Pnma (POSCAR.III, symmetrized via
   1.00000000000000
     8.7032434407543473    0.0000000000000000    0.0000000000000000
     0.0000000000000000    2.9349047670972270    0.0000000000000000
     0.0000000000000000    0.0000000000000000    9.4892012168552959
   Al   C
    16    12
Direct
  0.7372093599188929  0.2500000000000000  0.7136306477819190
  0.7627906400811071  0.7500000000000000  0.2136306477819190
  0.2627906400811071  0.7500000000000000  0.2863693522180810
  0.2372093599188929  0.2500000000000000  0.7863693522180810
  0.6063043166502808  0.2500000000000000  0.4707003951628863
  0.8936956833497192  0.7500000000000000  0.9707003951628863
  0.3936956833497192  0.7500000000000000  0.5292996048371137
  0.1063043166502808  0.2500000000000000  0.0292996048371137
  0.1374957759450339  0.2500000000000000  0.5177890409486778
  0.3625042240549661  0.7500000000000000  0.0177890409486778
  0.8625042240549661  0.7500000000000000  0.4822109590513222
  0.6374957759450339  0.2500000000000000  0.9822109590513222
  0.9932809183437712  0.2500000000000000  0.2795179626542890
  0.5067190816562288  0.7500000000000000  0.7795179626542890
  0.0067190816562288  0.7500000000000000  0.7204820373457110
  0.4932809183437712  0.2500000000000000  0.2204820373457110
  0.9285470810138605  0.2500000000000000  0.6187220005827854
  0.5714529189861395  0.7500000000000000  0.1187220005827854
  0.0714529189861395  0.7500000000000000  0.3812779994172146
  0.4285470810138605  0.2500000000000000  0.8812779994172146
  0.3911565475003371  0.2500000000000000  0.4018945523721129
  0.1088434524996629  0.7500000000000000  0.9018945523721058
  0.6088434524996629  0.7500000000000000  0.5981054476278942
  0.8911565475003371  0.2500000000000000  0.0981054476278942
  0.7744048364141989  0.2500000000000000  0.3539263382519593
  0.7255951635858011  0.7500000000000000  0.8539263382519593
  0.2255951635858011  0.7500000000000000  0.6460736617480407
  0.2744048364141989  0.2500000000000000  0.1460736617480407

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

```
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

anti-$CaFe_2O_4$ – 0 GPa

```
Al4C3 anti-CaFe2O4-type, volume pre-scal
   1.00000000000000
     2.9852194567723527    0.0000000000000000    0.0000000000000000
     0.0000000000000000    9.1198705024020921    0.0000000000000000
     0.0000000000000000    0.0000000000000000   10.5792008400718061
   Al   C
    16    12
Direct
  0.2500000000000000  0.4144072947002274  0.5809008803943491
  0.7500000000000000  0.5855927052997728  0.4190991196056512
  0.2500000000000000  0.9144072947002272  0.9190991196056509
  0.7500000000000000  0.0855927052997727  0.0809008803943488
  0.2500000000000000  0.4889137898529304  0.2054812901082441
  0.7500000000000000  0.5110862101470698  0.7945187098917560
  0.2500000000000000  0.9889137898529302  0.2945187098917558
  0.7500000000000000  0.0110862101470698  0.7054812901082440
  0.2500000000000000  0.1351843018774951  0.5212377979060971
  0.7500000000000000  0.8648156981224979  0.4787622020939027
  0.2500000000000000  0.6351843018775021  0.9787622020939029
  0.7500000000000000  0.3648156981225048  0.0212377979060973
  0.7500000000000000  0.2770966780917617  0.3071244884571125
  0.2500000000000000  0.7229033219082382  0.6928755115428877
  0.2500000000000000  0.2229033219082383  0.8071244884571123
  0.7500000000000000  0.7770966780917618  0.1928755115428875
  0.7500000000000000  0.7411535668307502  0.8617177029566370
  0.2500000000000000  0.2588464331692499  0.1382822970433632
  0.7500000000000000  0.2411535668307501  0.6382822970433630
  0.2500000000000000  0.7588464331692498  0.3617177029566369
  0.2500000000000000  0.9221420520041845  0.6119178216534967
  0.7500000000000000  0.5778579479958155  0.1119178216534966
  0.2500000000000000  0.4221420520041844  0.8880821783465033
  0.7500000000000000  0.0778579479958156  0.3880821783465034
  0.7500000000000000  0.0805052220197591  0.8841128946722824
  0.2500000000000000  0.9194947779802410  0.1158871053277176
  0.7500000000000000  0.5805052220197590  0.6158871053277176
  0.2500000000000000  0.4194947779802408  0.3841128946722825

  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

```
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

anti-$CaFe_2O_4$ – 50 GPa

```
Al4C3 anti-CaFe2O4-type, volume pre-scal
   1.00000000000000
     2.7900451726561628    0.0000000000000000    0.0000000000000000
     0.0000000000000000    8.5236114258706834    0.0000000000000000
     0.0000000000000000    0.0000000000000000    9.8875304351378670
   Al   C
    16    12
Direct
  0.2500000000000000  0.4245267340863990  0.5873716912218541
  0.7500000000000000  0.5754732659136010  0.4126283087781459
  0.2500000000000000  0.9245267340863990  0.9126283087781459
  0.7500000000000000  0.0754732659136010  0.0873716912218541
  0.2500000000000000  0.4800601400098898  0.2061656728190329
  0.7500000000000000  0.5199398599901102  0.7938343271809671
  0.2500000000000000  0.9800601400098898  0.2938343271809671
  0.7500000000000000  0.0199398599901102  0.7061656728190329
  0.2500000000000000  0.1484441334435687  0.5143475653135141
  0.7500000000000000  0.8515558665564242  0.4856524346864859
  0.2500000000000000  0.6484441334435758  0.9856524346864859
  0.7500000000000000  0.3515558665564313  0.0143475653135141
  0.7500000000000000  0.2662336654069293  0.3000390016478320
  0.2500000000000000  0.7337663345930707  0.6999609983521680
  0.2500000000000000  0.2337663345930707  0.8000390016478320
  0.7500000000000000  0.7662336654069293  0.1999609983521680
  0.7500000000000000  0.7536300515530598  0.8590001755942538
  0.2500000000000000  0.2463699484469402  0.1409998244057462
  0.7500000000000000  0.2536300515530598  0.6409998244057462
  0.2500000000000000  0.7463699484469402  0.3590001755942538
  0.2500000000000000  0.9301796991213749  0.6107702309581455
  0.7500000000000000  0.5698203008786251  0.1107702309581455
  0.2500000000000000  0.4301796991213749  0.8892297690418545
  0.7500000000000000  0.0698203008786251  0.3892297690418545
  0.7500000000000000  0.0968949186839723  0.8893454137177415
```

```
  0.2500000000000000  0.9031050813160277  0.1106545862822585
  0.7500000000000000  0.5968949186839723  0.6106545862822585
  0.2500000000000000  0.4031050813160277  0.3893454137177415

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

I-43d – 0 GPa

```
Al4C3 I-43d polymorph (findsym-output, f
   1.00000000000000
     6.5463773857702332    0.0000000000000000    0.0000000000000000
     0.0000000000000000    6.5463773857702332    0.0000000000000000
     0.0000000000000000    0.0000000000000000    6.5463773857702332
   Al   C
    16    12
Direct
  0.2990010309058072  0.2990010309058072  0.2990010309058072
  0.2009989690941928  0.7009989690941930  0.7990010309058070
  0.7009989690941930  0.7990010309058070  0.2009989690941928
  0.7990010309058070  0.2009989690941928  0.7009989690941930
  0.5490010309058070  0.5490010309058070  0.5490010309058070
  0.9509989690941930  0.4509989690941928  0.0490010309058072
  0.0490010309058072  0.9509989690941930  0.4509989690941928
  0.4509989690941928  0.0490010309058072  0.9509989690941930
  0.7990010309058070  0.7990010309058070  0.7990010309058070
  0.7009989690941930  0.2009989690941928  0.2990010309058072
  0.2009989690941928  0.2990010309058072  0.7009989690941930
  0.2990010309058072  0.7009989690941930  0.2009989690941928
  0.0490010309058072  0.0490010309058072  0.0490010309058072
  0.4509989690941928  0.9509989690941930  0.5490010309058070
  0.5490010309058070  0.4509989690941928  0.9509989690941930
  0.9509989690941930  0.5490010309058070  0.4509989690941928
  0.3750000000000000 -0.0000000000000000  0.2500000000000000
  0.1250000000000000 -0.0000000000000000  0.7500000000000000
  0.6250000000000000  0.5000000000000000  0.2500000000000000
  0.8750000000000000  0.5000000000000000  0.7500000000000000
```

```
  0.2500000000000000  0.3750000000000000 -0.0000000000000000
  0.7500000000000000  0.1250000000000000 -0.0000000000000000
  0.2500000000000000  0.6250000000000000  0.5000000000000000
  0.7500000000000000  0.8750000000000000  0.5000000000000000
 -0.0000000000000000  0.2500000000000000  0.3750000000000000
 -0.0000000000000000  0.7500000000000000  0.1250000000000000
  0.5000000000000000  0.2500000000000000  0.6250000000000000
  0.5000000000000000  0.7500000000000000  0.8750000000000000

  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
  0.00000000E+00  0.00000000E+00  0.00000000E+00
```

I-43d – 50 GPa

```
Al4C3 I-43d polymorph (findsym-output, f
   1.00000000000000
     6.0865525100003293    0.0000000000000000    0.0000000000000000
     0.0000000000000000    6.0865525100003293    0.0000000000000000
     0.0000000000000000    0.0000000000000000    6.0865525100003293
   Al   C
    16    12
Direct
  0.3135103407318738  0.3135103407318738  0.3135103407318738
  0.1864896592681262  0.6864896592681262  0.8135103407318738
  0.6864896592681262  0.8135103407318738  0.1864896592681262
  0.8135103407318738  0.1864896592681262  0.6864896592681262
  0.5635103407318738  0.5635103407318738  0.5635103407318738
  0.9364896592681262  0.4364896592681262  0.0635103407318738
  0.0635103407318738  0.9364896592681262  0.4364896592681262
  0.4364896592681262  0.0635103407318738  0.9364896592681262
  0.8135103407318738  0.8135103407318738  0.8135103407318738
  0.6864896592681262  0.1864896592681262  0.3135103407318738
  0.1864896592681262  0.3135103407318738  0.6864896592681262
  0.3135103407318738  0.6864896592681262  0.1864896592681262
  0.0635103407318738  0.0635103407318738  0.0635103407318738
  0.4364896592681262  0.9364896592681262  0.5635103407318738
  0.5635103407318738  0.4364896592681262  0.9364896592681262
```

```
0.9364896592681262  0.5635103407318738  0.4364896592681262
0.3750000000000000  0.0000000000000000  0.2500000000000000
0.1250000000000000  0.0000000000000000  0.7500000000000000
0.6250000000000000  0.5000000000000000  0.2500000000000000
0.8750000000000000  0.5000000000000000  0.7500000000000000
0.2500000000000000  0.3750000000000000  0.0000000000000000
0.7500000000000000  0.1250000000000000  0.0000000000000000
0.2500000000000000  0.6250000000000000  0.5000000000000000
0.7500000000000000  0.8750000000000000  0.5000000000000000
0.0000000000000000  0.2500000000000000  0.3750000000000000
0.0000000000000000  0.7500000000000000  0.1250000000000000
0.5000000000000000  0.2500000000000000  0.6250000000000000
0.5000000000000000  0.7500000000000000  0.8750000000000000

0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
0.00000000E+00  0.00000000E+00  0.00000000E+00
```

5 État de la campagne

Polymorphe	Points préparés	Amorçage	État
Pnma-VII, Pbam, Pnma, R-3m, Pnma-III, anti-$CaFe_2O_4$, I-43d	11/11 (0–100 GPa)	WAVECAR PBE, même pression	fichiers prêts, non soumis
Fd-3m	11/11 (0–100 GPa)	WAVECAR PBE à $P = 0$ seulement, chaîne ascendante ensuite	**soumis** (job SLURM 58461)

Les fichiers des sept premiers polymorphes sont générés dans `hp_curves_r2SCAN/` (`gen_hp_runs_r2scan.py`) mais pas encore soumis à SLURM. La chaîne Fd-3m a été soumise séparément (`gen_hp_run_fd3m_r2scan.py`), le point $P = 0$ étant le seul à disposer d'un `WAVECAR` PBE convergé au moment du lancement (le scan PBE de Fd-3m, lancé en chaîne ascendante unique lui aussi, est toujours en cours – job SLURM 58460).

6 Prochaines étapes

(i) Soumettre les sept chaînes r2SCAN restantes (`hp_curves_r2SCAN/submit_chain.sh`), après revue.

(ii) Une fois les points r2SCAN convergés, extraire E, V, H (`hp_curves_r2SCAN/extract_hp.py`, même format que le scan PBE) et construire les diagrammes $V(P)$, $E(P)$, $H(P)$ r2SCAN, superposés aux courbes PBE du rapport principal.

(iii) Comparer les pressions de transition PBE (26 et 52 GPa) aux valeurs r2SCAN correspondantes, avec la même analyse de précision ($\delta P_t \approx \sqrt{2}\,\delta H/|\Delta V(P_t)|$) que dans le rapport principal.

(iv) Vérifier en particulier si r2SCAN lève ou confirme la quasi-dégénérescence Pbam $\leftrightarrow$ Pnma-III observée en PBE au-delà de 70 GPa.

A Références du dépôt

Scripts de génération : `hp_curves/gen_hp_runs_r2scan.py` (sept polymorphes convergés en PBE), `hp_curves/gen_hp_run_fd3m_r2scan.py` (Fd-3m, chaîne ascendante). Données d'entrée et de sortie dans `hp_curves_r2SCAN/Al4C3_<polymorphe>/P<pression>/`, script d'extraction `hp_curves_r2SCAN/extract_hp.py`. Scan PBE de référence : `rapport_activite_Al4C3_HP_scan.tex` (commit `effe450`).